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# Decomposing a graph into forests and a matching

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ABSTRACT

The fractional arboricity of a graph $G$, denoted by $\gamma_f(G)$, is defined as $\gamma_f(G) = \max_{H \subseteq G, v(H)>1} \frac{e(H)}{v(H)-1}$. The famous Nash-Williams' Theorem states that a graph $G$ can be partitioned into at most $k$ forests if and only if $\gamma_f(G) \leq k$. A graph is $d$-bounded if it has maximum degree at most $d$. The Nine Dragon Tree (NDT) Conjecture [posed by Montassier, Ossona de Mendez, Raspaud, and Zhu, at [11]] asserts that if $\gamma_f(G) \leq k + \frac{d}{k+d+1}$, then $G$ decomposes into $k+1$ forests with one being $d$-bounded. In this paper, it is proven that the Nine Dragon Tree Conjecture is true for all the cases in which $d = 1$.

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## 1. Introduction

Throughout this paper, graphs are finite. A graph may have parallel edges but no loops. Given a graph $G$, $V(G)$ and $E(G)$ denote the vertex set and the edge set of $G$, respectively. Let $v(G) := |V(G)|$ and $e(G) := |E(G)|$. If $X \subseteq V(G)$, then $G[X]$ is the subgraph of $G$ induced by $X$; $G - X$ is the graph obtained from $G$ by deleting vertices

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in $X$. If $S$ is a set of edges, the edge-induced subgraph $G[S]$ is the subgraph of $G$ whose edge set is $S$ and whose vertex set consists of all ends of edges of $S$. Let $N_G(v)$ denote the neighbors of $v$, $E_G(v)$ denote the edges that are incident to $v$, and let $d_G(v)$ be the degree of $v$, i.e., $d_G(v) = |E_G(v)|$.

Given a graph $G$, a *decomposition* of a graph $G$ consists of edge-disjoint subgraphs with union $G$. The *arboricity* of $G$, denoted $\gamma(G)$, is the minimum number of forests needed to decompose it. The *fractional arboricity* of $G$ is defined as

$$\gamma_f(G) = \max_{H \subseteq G, v(H)>1} \frac{e(H)}{v(H)-1}.$$

This notion was introduced by Payan [14]; see also [2,6]. The famous Nash-Williams' Theorem states a necessary and sufficient condition for $\gamma(G) \leq k$:

**Theorem 1.1** *(Nash-Williams 1964 [13]). A graph $G = (V, E)$ can be decomposed into at most $k$ forests if and only if $\gamma_f(G) \leq k$.*

This is a sparseness condition. Say a graph is *$d$-bounded* if it has maximum degree at most $d$. Montassier et al. [11] posed *the Nine Dragon Tree (NDT) Conjecture* (honoring a famous tree in Kaohsiung, Taiwan that is far from acyclic):

**Conjecture 1.2** *(The NDT (Nine Dragon Tree) Conjecture [11]). Suppose $G$ is a graph and $k$, $d$ are non-negative integers. If $\gamma_f(G) \leq k + \frac{d}{k+d+1}$, then $G$ decomposes into $k+1$ forests with one being $d$-bounded.*

Let a $(k, d)$-decomposition of a graph $G$ be a decomposition of $G$ into $k+1$ forests with one having maximum degree at most $d$. Graphs having such a decomposition are $(k, d)$-decomposable. Regarding the progress of Nine Dragon Tree (NDT) Conjecture, Montassier et al. [11] proved the cases of $(k, d) = (1, 1)$ and $(k, d) = (1, 2)$, and they showed that no larger value of $\gamma_f(G)$ is sufficient. Kim et al. [10] proved the cases of $(k, d) = (k, k+1)$ and $(k, d) = (1, d)$ for $d \leq 6$. Chen et al. [3] proved the cases $(k, d)$ for all $d$ when $k \leq 2$, except for $(k, d) = (2, 1)$. By using an extension of the matroid intersection theorem [4] due to Aharoni and Berger [1], Kaiser et al. in [9] proved that, if $\gamma_f(G) \leq k + \frac{1}{3k+2}$, then $G$ is $(k, 1)$-decomposable.

In this paper, we shall prove that, if $\gamma_f(G) \leq k + \frac{1}{k+2}$, then $G$ is $(k, 1)$-decomposable. By using the methodology that is first developed in this paper, a proof of the NDT Conjecture has been obtained by Jiang and Yang in [8]. We expect there will be more applications of these ideas. The presentation of the proof here is more algorithmic, while the proof in [8] was presented by using partially ordered sets.

A slightly different sparseness condition places a bound on the average vertex degree in all subgraphs. The *maximum average degree* of a graph $G$, denoted by $\mathrm{mad}(G)$, is defined as

$$\mathrm{mad}(G) = \max_{H \subseteq G} \frac{2e(H)}{v(H)}.$$

It has already attracted a lot of attention and has a lot of applications. We compare $\mathrm{mad}(G)$ and $\gamma_f(G)$. Always $\mathrm{mad}(G) < 2\gamma_f(G)$, but the conditions $\mathrm{mad}(G) < 2a$ and $\gamma_f(G) \leq a$ are not equivalent.

Montassier et al. [12] posed the question of finding the weakest bound on $\mathrm{mad}(G)$ to guarantee decomposition into one forest and a $d$-bounded graph. They proved that $\mathrm{mad}(G) < 4 - \frac{8d+12}{d2+6d+6}$ is sufficient and that $\mathrm{mad}(G) = 4 - \frac{4}{d+2}$ is not (seen by subdividing every edge of a $(2d+2)$-regular graph).

Next we pose a result analogous (in some sense) to the Nine Dragon Tree Conjecture. Suppose that $\sigma$ is an orientation of $G$, we use $G^\sigma$ to denote the oriented graph. Let $\Delta^+(G^\sigma) = \max_{v \in V} d^+_{G^\sigma}(v)$, where $d^+_{G^\sigma}(v)$ is the outdegree of $v$ in $G^\sigma$. The following result is well-known (cf. [7], Theorem 4).

**Theorem 1.3.** *Let $G$ be a graph. Then $G$ has an orientation $\sigma$ such that $\Delta^+(G^\sigma) \leq k$ if and only if $\mathrm{mad}(G) \leq 2k$.*

A graph $G$ is called a *pseudoforest* if there is an orientation $\sigma$ of $G$ such that $\Delta^+(G^\sigma) \leq 1$, i.e., $G$ is a graph whose every component has at most one cycle. The following corollary comes from Theorem 1.3.

**Corollary 1.4.** *For any graph $G$, if $\mathrm{mad}(G) \leq 2k$, then $G$ decomposes into $k$ pseudoforests.*

The following result is given by Fan et al. in [5], and is, in a certain sense, analogous to the Nine Dragon Tree Conjecture.

**Theorem 1.5** *([5]). For any graph $G$, if $\mathrm{mad}(G) \leq 2k + \frac{2d}{k+d+1}$, then $G$ decomposes into $k+1$ pseudoforests with one being $d$-bounded.*

The following is our main result, which implies that the Nine Dragon Tree Conjecture is true for all the cases in which $d = 1$.

**Theorem 1.6.** *For any non-negative integer $k$, if for any subgraph $H$ of $G$,*

$$e(H) \leq k(v(H) - 1) + \frac{1}{k+2} v(H),$$

*then $G$ decomposes into $k$ forests and a matching.*

In Section 2, we shall prove Theorem 1.6. In fact, the methods we use here further developed the methodology that was used in the proof of Theorem 1.5 in [5].

## 2. Proof of Theorem 1.6

We argue by contradiction to prove Theorem 1.6. If $k = 0$, then it is easy to show that Theorem 1.6 holds true. Suppose Theorem 1.6 is not true, then $k \geq 1$; by Nash-Williams' Theorem 1.1, we have $\gamma_f(G) > k$. Let $G$ be a minimal counterexample of Theorem 1.6 for such a $k$. Note that the condition of Theorem 1.6 implies that $\gamma_f(G) < k + 1$.

Let $\mathcal{F}_{k+1}$ be the set of all $(k + 1)$-tuples $\mathbb{F} = (F_1, \ldots, F_k, F_{k+1})$ of edge-disjoint spanning forests in $G$. By Theorem 1.1, $G$ decomposes into $k + 1$ forests, let $\mathbb{F} = (F_1, \ldots, F_k, F_{k+1})$ be such a decomposition of $G$. Since $\gamma_f(G) > k$, $E(F_{k+1}) \neq \emptyset$.

**Definition 2.1.** A forest decomposition $\mathbb{F} = (F_1, F_2, \ldots, F_k, F_{k+1})$ of $G$ is an *optimal* $(k + 1)$*-forest decomposition* if

(1) $|E(F_{k+1})|$ is minimum among all $(k + 1)$-forest decompositions of $G$;
(2) subject to (1), $\sum_{v \in V(G)} \max\{0, d_{F_{k+1}}(v) - 1\}$ is minimum.

First, we shall prove the following result; it is named as a theorem, since some slight modifications of the proof will lead to a new proof of Nash-Williams' Theorem 1.1 (refer [15]).

**Theorem 2.2.** *Assume $\mathbb{F} = (F_1, \ldots, F_k, F_{k+1})$ is an optimal $(k + 1)$-forest decomposition of $G$, then $F_1, \ldots, F_k$ are $k$ spanning trees of $G$.*

**Proof.** Assume $\mathbb{F}$ is an optimal $(k + 1)$-forest decomposition of $G$, define $F = F_1 \cup F_2 \cup \ldots \cup F_k$; then the number of edges in $F$ is maximum (for all such $k$-forest unions). *Color the edges in $F$ blue and the edges in $G - F$ red.* Since $G$ is a counterexample of Theorem 1.6, there is at least a vertex $s$ in $F_{k+1}$ with $d_{F_{k+1}}(s) = d_{G-F}(s) > 1$.

For $i = 1, 2, \ldots, k$, let $T_i$ be the component of $F_i$ containing vertex $s$. Orient each of the trees $T_i$ as rooted at $s$ (so $s$ is a sink and each other vertex in $T_i$ has outdegree 1), and let $F' = T_1 \cup \ldots \cup T_k$. (The other edges of $F$ and red edges are not oriented). Even though $T_i$s are just subtrees (at this time), we call each $T_i$ a tree for convenience.

We define an *extending path* $P$ in $G$ recursively ($P$ is required to be a path in the whole process) as follows:

- A red-edge-path starting from $s$ is an extending path.
- If $P$ is an extending path ending at a red edge, then add to the end of $P$ a forward directed $F'$-path is an extending path.
- If $P$ is an extending path ending at an $F'$-edge, then add to the end of $P$ a red-edge-path is an extending path.

Assume $P$ is an extending path, and $u, v$ are vertices of $P$ with $u$ preceding $v$. If the sub-path $P[u, v]$ of $P$ from $u$ to $v$ contains a red edge, and there is a directed $F'$-path $Q$

from $u$ to $v$, then $Q$ is called a *short-cut* for $P$. Note that since $s$ is a sink for each $T_i$, no short cut of $P$ "covers" the initiating red edge of $P$. An extending path $P$ is *optimal* if it has no short-cuts.

We say a vertex $x$ (respectively an edge $e = xy$) is *extending-reachable* if $x = s$ or $x$ is the end of an extending path (respectively, $e$ is the end of an extending path). For an extending-reachable vertex $x$ or an edge $e$, there may be many different extending paths ending at $x$ or $e$.

**Lemma 2.3.** *Assume $e = xy$ is an extending-reachable blue edge, $P_0$ is an extending path ending at $e$, and $P_0$ has no short-cuts. Let $e' = wv$ be the last red edge on $P_0$. Then $\bigcup_1^k E(T_i) - e + e'$ has a $k$-forest decomposition $T'_1, T'_2, \ldots, T'_k$ such that each $T'_i$ is still a subtree of $G$ with $s \in V(T'_i)$.*

**Proof.** Suppose that $P := P_0[v, y] = x_0x_1 \ldots x_{t-1}x_t$ being a blue directed path from $x_0 = v$ (and $x_{-1} = w$) to $x_{t-1} = x$ (and $x_t = y$). A *segment* of $P$ is defined to be a maximal subpath of $P$ contained in some $T_i$ $(1 \le i \le k)$.

Assume the segments of $P$ are $P_1 = (x_0, x_1, \ldots, x_{a_1})$, $P_2 = (x_{a_1}, x_{a_1+1}, \ldots, x_{a_2}), \ldots,$ $P_l = (x_{a_{(l-1)}}, \ldots, x_{a_l})$ (where $x_{a_l} = x_t$, $x_{a_l-1} = x_{t-1}$), and $P_j$ $(1 \le j \le l)$ is a path in the blue tree $T_{b_j}$. Define the *label* $\mu(P)$ of $P$ as $\mu(P) = (a_1, \ldots, a_{(l-1)})$.

Assume that $P = (x_0, x_1, \ldots, x_t)$ is a blue directed path with initial vertex $v = x_0$ and the last directed edge $(x, y)$ such that the number of segments in $P$ is minimum among all the blue paths with initial vertex $v$ and the last directed edge $(x, y)$; and subject to above, $P$ is chosen with minimal $\mu(P)$ (according to the lexicographic order of integer sequences).

We claim that for $j = 2, \ldots, l$, $x_{a_{(j-1)}}$ is the first vertex in $P$ for which there is a directed path in $T_{b_j}$ to $x_{a_j-1}$ (where $x_{a_j-1}$ is the previous vertex of $x_{a_j}$ in the segment $P_j$).

Check this in turn from $P_2$ to $P_l$. Suppose otherwise, $j$ is the first index such that this claim does not hold true, then there is a directed path $P'$ in $T_{b_j}$ from $x_i$ to $x_{a_j-1}$ for some $i < a_{(j-1)}$. We replace the subpath $P[x_i, x_{a_j-1}]$ of $P$ with $P'$, suppose the obtained blue path is $P''$. Then, either the number of segments of $P''$ is smaller than $P$, or the number of segments of $P''$ equals $P$ but $\mu(P'') \prec \mu(P)$. This contradicts the assumptions of $P$.

For $j = 0, 1, \ldots, l$, let $e_j = x_{a_j-1}x_{a_j}$ (where $a_0 = 0$, $x_0 = v$, and $x_{-1} = w$). For $j = l, l-1, \ldots, 1$, if $T_{b_j} + e_{j-1}$ is still a tree, then let $T_{b_j} := T_{b_j} + e_{j-1}$; otherwise, let $T_{b_j} := T_{b_j} - e_j + e_{j-1}$. (Read these as sentences in an algorithm, the resulting tree is still being named $T_{b_j}$. We use the same name for the new tree, because we may need to operate on the same tree more than once. This happens when $b_j = b_{j'}$ for some $j > j'$.)

We claim that the new $T_{b_j}$ $(1 \le j \le l)$ is still a tree. It suffices to show that if $T_{b_j} + e_{j-1}$ contains a cycle, then the unique cycle in $T_{b_j} + e_{j-1}$ contains the edge $e_j$. This is true for $j \ge 2$; since otherwise, there would be a directed path in $T_{b_j}$ connecting $x_{a_{(j-1)}-1}$ to $x_{a_j-1}$, contrary to the claim that $x_{a_{(j-1)}}$ is the first vertex in $P$ for which

there is a directed path in $T_{b_j}$ to $x_{a_j-1}$. For $j = 1$, $e_0 = wv$; this is true, since there is no blue directed path from $w$ to $x$ (since $P_0$ has no short-cuts). Thus $T_{b_j} - e_j + e_{j-1}$ is a tree.

If $j' < j$ and $b_{j'} = b_j$, then $x_{a_{(j'-1)}}$ is still the first vertex in $P$ for which there is a directed path in $T'_{b_j}$ to $x_{a_{j'}-1}$. This is because the segments $P_j$ and $P_{j'}$ are disjoint.

Let $T'_{b_j}$ be the new tree obtained from the above operation. The orientation of $T'_{b_j}$ (as a rooted tree with root $s$) is obtained by the following steps: first, reversing the orientations of edges in the blue directed path $(x_{a_{(j-1)}}, \ldots, x_{a_j-1})$, i.e., changing it to a directed path $(x_{a_j-1}, \ldots, x_{a_{(j-1)}})$; then reorienting the arc $(x_{a_{(j-1)}-1}, x_{a_{(j-1)}})$ as $(x_{a_{(j-1)}}, x_{a_{(j-1)}-1})$ (for $j \geq 2$), orienting the (new blue) edge $x_0x_{-1} = vw$ as $(v, w)$ (for $j = 1$).

Denote by $T'_1, T'_2, \ldots, T'_k$ the trees obtained from $T_1, T_2, \ldots, T_k$ by the operations above, note that each $T'_i$ is still a subtree of $G$ with $s \in V(T'_i)$, this proves the lemma. □

**Claim 2.4.** *For any $i \in \{1, 2, \ldots, k\}$, all extending-reachable vertices are contained in $V(T_i)$.*

**Proof.** Let $P$ be an extending path ending at edge $e = xy$, and assume every other vertex of $P$ is contained in $V(T_i)$, but $y$ is not contained in $V(T_i)$. Without loss of generality, we may assume that $P$ has no short-cuts.

If $e$ is a red edge, then simply add $e$ to $F_i$. If $e \in T_j$, then add $e$ to $F_i$; suppose $e'$ is the last red edge of $P$; by Lemma 2.3, $\bigcup_1^k E(T_i) - e + e'$ has a $k$-forest decomposition; then replace the trees $T_1, T_2, \ldots, T_k$ by the corresponding trees obtained in Lemma 2.3. In any case, we obtain another $k$-forest union with one more edge, contrary to the optimality of $\mathbb{F}$. □

Let $U$ be the set of all extending-reachable vertices. It follows from Claim 2.4 that $U \subseteq V(T_i)$, for all $i \in \{1, 2, \ldots, k\}$.

**Claim 2.5.** *For any $i \in \{1, 2, \ldots, k\}$, $T_i[U]$ is connected. For each component $C$ of $F_{k+1}$, either $C \cap U = \emptyset$ or $C \subseteq U$.*

**Proof.** Suppose otherwise; $C_1$ and $C_2$ are disjoint components of $T_i[U]$, $s \in C_1$ and $u \in C_2$. Considering the directed path $P$ in $T_i$ from $u$ to $s$, since $C_1$ and $C_2$ are disconnected in $T_i[U]$, there is a vertex $w \in P$, such that $w \notin U$; suppose $w$ is the first such vertex in $P$ from $u$ to $s$; but then vertex $u \in U$ and $P$ witness that $w$ is extending-reachable. This is a contradiction.

The claim of, for each component $C$ of $F_{k+1}$ either $C \cap U = \emptyset$ or $C \subseteq U$, comes from the definition of extending paths. □

If $U \subsetneq V(G)$, by the minimality of $G$, $G[U]$ can be decomposed into $k + 1$ forests, with the last forest of maximum degree at most 1. By Claim 2.5, for $1 \leq i \leq k + 1$, replacing in $F_i$ the edges of $F_i[U]$ with the edges of the $i$th forest in the decomposition of $G[U]$, we obtain a decomposition $\mathbb{F}'' = (F''_1, \ldots, F''_k, F''_{k+1})$ of $G$ such that

$\sum_{v\in V(G)} \max\{0, d_{F''_{k+1}}(v) - 1\}$ is decreased (since $\max\{0, d_{F''_{k+1}}(s) - 1\}$ is decreased), contrary to the optimality of $\mathbb{F}$. We concluded that $U = V(G) = V(T_i)$, for each $i \in \{1, 2, \ldots, k\}$; this proves Theorem 2.2. $\square$

Suppose that $\mathbb{F} = (T_1, \ldots, T_k, F_{k+1})$ is an optimal decomposition of $G$. Then by Theorem 2.2, $T_1, \ldots, T_k$ are $k$ spanning trees of $G$. Since $G$ is a counterexample of Theorem 1.6, there is a vertex s in $F_{k+1}$ with $d_{F_{k+1}}(s) > 1$. Fix vertex $s$ such that $d_{F_{k+1}}(s) > 1$, orient each of the trees $T_i$ $(1 \le i \le k)$ and forest $F_{k+1}$ as rooted at $s$ (so $s$ is a sink and each other vertex in $T_i$ has outdegree 1); we call $\sigma$ *an optimal orientation of* $G$; denote the oriented graph $G$ as $G^\sigma$, the oriented $T_1, \ldots, T_k$ and $F_{k+1}$ as $T_1^\sigma, \ldots, T_k^\sigma$ and $F_{k+1}^\sigma$, respectively. Then *color the arcs in* $T_1, \ldots, T_k$ *blue and the arcs in* $F_{k+1}$ *red.* We call such a coloring *a red–blue-coloring (associated with* $\sigma$*) of* $G^\sigma$.

For $v \in V(G)$, let $N_i(v)$ denote the neighbors of $v$ in $T_i^\sigma$, $d_i(v)$ denote the degree of $v$ in $T_i^\sigma$, $N_i[v] = N_i(v) \cup \{v\}$. The subgraph induced by all red (blue) arcs is denoted by $G_r^\sigma$ ($G_b^\sigma$, respectively), undirected graph $G_r$ ($G_b$, respectively) is obtained from $G_r^\sigma$ ($G_b^\sigma$, respectively) by ignoring the directions of the arcs in the digraph. Let $N_r(v)$ and $N_b(v)$ denote the neighbors of $v$ in $G_r$ and $G_b$, $N_r[v] = N_r(v) \cup \{v\}$, $N_b[v] = N_b(v) \cup \{v\}$, $N_r^+(v)$ and $N_b^+(v)$ denote the outneighbors of $v$ in $G_r^\sigma$ and $G_b^\sigma$, $E_r(v)$ denote the edges that are incident to $v$ in $G_r^\sigma$, let $d_r^+(v)$ and $d_b^+(v)$ be the outdegree of $v$ in $G_r^\sigma$ and $G_b^\sigma$, respectively. Notations $d_r(v)$, $d_b(v)$, $\Delta_r^+(G^\sigma)$, $\Delta_b^+(G^\sigma)$, $\Delta_r(G^\sigma)$, and $\Delta_b(G^\sigma)$ are defined similarly.

Suppose that $\sigma$ is an optimal orientation of $G$. For any vertex $x \in V(G)$ and $x \neq s$, there is an unique directed path from $x$ to $s$ in $T_i^\sigma$, denote this path as $xT_i^\sigma s$. For $u \in V(G)$, let $T_r^u$ denote the (red) component of $F_{k+1}^\sigma$ that contains $u$; note that for any $v \in V(T_r^u)$, $T_r^v = T_r^u$. For each vertex $v \in V(G)$, if $d_r(v) < 1$, $v$ is called *an unsaturated vertex*; otherwise, $v$ is called *a saturated vertex.*

Next, we shall design the **Growing Algorithm**. The algorithm will use an optimal orientation $\sigma$; $\sigma$ may get updated as the algorithm is running (in other words, it is dynamic during the algorithm running). By picking some vertices in the dynamic $G^\sigma$, the algorithm will construct a vertex set $P$. By using $P$, the algorithm constructs a digraph $\vec{H} \subseteq G^\sigma$ that does not satisfy the condition of Theorem 1.6. To facilitate the analysis, for each vertex $x \in P$, a function $f(x)$ is tracked in the algorithm (for understanding, $f(x)$ can be interpreted as *the father of* $x$).

**Initialization of the algorithm**:
Define $j := 0$; $\Gamma_0 := \{\sigma : \sigma \text{ is an optimal orientation of } G, \text{ and } d_r(s) > 1, \text{ for fixed } s\}$;
pick $\sigma_0 \in \Gamma_0$, such that $T_r^s$ in $G^{\sigma_0}$ is minimal among all $\sigma \in \Gamma_0$; let $\sigma := \sigma_0$;
$P := \{s\}$; $f(s) := null$;
define $\vec{H} \subseteq G^\sigma$, such that $E(\vec{H}) := E(T_r^s) \bigcup_{x\in V(T_r^s)} E_b^+(x)$, $\vec{H} := G^\sigma[E(\vec{H})]$;
$\Gamma_{j+1} := \{\sigma' : \sigma' \text{ is an optimal orientation of } G, \text{ s.t. } \vec{H} \subseteq G^{\sigma'}\}$; $j := j + 1$;

**while** there exists $u \in P$ and $u$ is unmarked, do

**(a)** for each $v \in V(T_r^u)$,
for each $w \in N_b^+(v)$,
pick $\sigma_j \in \Gamma_j$, such that $T_r^w$ in $G^{\sigma_j}$ is minimal among all $\sigma \in \Gamma_j$; $\sigma := \sigma_j$;
**(i)** if $w$ is a saturated vertex, and $V(T_r^w) \cap P = \emptyset$, then
$P := P \cup \{w\}$, $f(w) := u$;
$E(\vec{H}) := E(\vec{H}) \cup E(T_r^w) \bigcup_{x \in V(T_r^w)} E_b^+(x)$, $\vec{H} := G^\sigma[E(\vec{H})]$;
$\Gamma_{j+1} := \{\sigma' : \sigma'$ is an optimal orientation of $G$, s.t. $\vec{H} \subseteq G^{\sigma'}\}$;
$j := j + 1$;
**(ii)** else if $w$ is an unsaturated vertex, then
$E(\vec{H}) := E(\vec{H}) \cup E_b^+(w)$, $\vec{H} := G^\sigma[E(\vec{H})]$;
for every $q \in N_b^+(w)$,
pick $\sigma_j \in \Gamma_j$, such that $T_r^q$ in $G^{\sigma_j}$ is minimal among all $\sigma \in \Gamma_j$; $\sigma := \sigma_j$;
if $q$ is a saturated vertex, and $V(T_r^q) \cap P = \emptyset$, then
$P := P \cup \{q\}$, $f(q) = u$;
$E(\vec{H}) := E(\vec{H}) \cup E(T_r^q) \bigcup_{x \in V(T_r^q)} E_b^+(x)$, $\vec{H} := G^\sigma[E(\vec{H})]$;
$\Gamma_{j+1} := \{\sigma' : \sigma'$ is an optimal orientation of $G$, s.t. $\vec{H} \subseteq G^{\sigma'}\}$;
$j := j + 1$;
**(b)** **mark** $u$ (for grown);

**end do;**

**return** $\Gamma_j$**,** $P$**,** $\vec{H}$**;**

Note that once a vertex has finished growing (i.e. it is marked), it will never grow again; since $G$ is a finite graph, the Growing Algorithm stops in finite steps. Also note that in the Growing Algorithm, the optimal orientation $\sigma$ might be updated; and once $\sigma$ is updated, the steps running next will be based on the updated $\sigma$ until the new update of $\sigma$. When the Growing Algorithm runs, it refers a sequence of optimal orientation set $\Gamma_j$; since subgraph $\vec{H} := G^\sigma[E(\vec{H})]$ of $G$ keeps on growing in the algorithm, by the definition of $\Gamma_j$, we have $\Gamma_0 \supseteq \Gamma_1 \supseteq ... \supseteq \Gamma_t$, where $\Gamma_t$ is the final one returned by the Growing Algorithm; since $\sigma \in \Gamma_t$, $\Gamma_t \neq \emptyset$.

The vertex set $P$ and subgraph $\vec{H}$ are dynamic in the algorithm, i.e., they may get updated as the algorithm runs. If there exist two unsaturated vertices $x'$ and $x''$, satisfying that $x'' \in N_b^+(x')$, $x', x'' \in V(G) \setminus V(\vec{H})$, and $T_r^{x'} \neq T_r^{x''}$, we say blue edge $x'x''$ is an *unsaturated (directed) blue edge with respect to* $\vec{H}$ *(in* $G^\sigma$*)*; for $v_0 \in V(\vec{H})$, if there exists a blue directed path $P_{v_0} := v_0 v_1 \dots v_t$ such that $v_{t-1} = x'$ and $v_t = x''$, we say $x'x''$ is *reachable from* $v_0$. In some cases, we would rather clearly indicate the subgraph $\vec{H}$ at a certain time of the algorithm; for this reason, we let $\vec{H}^u$ *denote the subgraph* $\vec{H}$ *that exists just before* $u$ *joined* $P$. The following definitions are aimed at vertices $u \in P$.

**Definition 2.6.** Suppose that vertex $u$ joined $P$ at time $j$, for any optimal orientation $\sigma \in \Gamma_{j+1}$, $T_r^u$ is the red subtree of $G^\sigma$ in the algorithm (and we always designate $u$ as

the root of $T_r^u$). We say that vertex $u$ has Property $P(u)$ with respect to $\vec{H}$, if both of the following are true:

(1) There is no unsaturated directed blue edge $x'x''$ with respect to $\vec{H}^u$ (in $G^\sigma$), such that $T_r^{x''} \neq T_r^u$, and $x'x''$ is reachable from any vertex $v \in V(T_r^u)$.
(2) for any red edge $vw \in E(T_r^u)$, there are no $i \in [k]$, such that $v' \in N_i^+(v) \setminus (V(\vec{H}^u) \cup V(T_r^u))$, $w' \in N_i^+(w) \setminus (V(\vec{H}^u) \cup V(T_r^u))$, and both $v'$ and $w'$ are unsaturated vertices.

Note that (1) of Property $P(u)$ implies that: for any $x' \in V(T_r^u)$, $x' \in V(G^\sigma) \setminus V(\vec{H}^u)$, if $x'$ is an unsaturated vertex, then for any $x'' \in N_b^+(x') \setminus V(\vec{H}^u)$ and $x'' \notin V(T_r^{x'})$, $x''$ is a saturated vertex. We call this $(1)'$ *of Property* $P(u)$.

For the following Fact 2.7, Lemma 2.8 and 2.9, *suppose that vertex $u$ joined $P$ at time $j$, for any optimal orientation $\sigma \in \Gamma_{j+1}$, $T_r^u$ is the red subtree of $G^\sigma$ with root $u$; and we assume that there is no unsaturated directed blue edge (with respect to $\vec{H}^u$) which is reachable from any vertex $y \in V(\vec{H}^u)$* (indeed, this will be part of the induction hypothesis in the inductive proof of Lemma 2.10).

**Fact 2.7.** *Assume there exists an unsaturated directed blue edge $x'x''$ (with respect to $\vec{H}^u$) which is reachable from $v_0 \in V(T_r^u)$, and $T_r^{x''} \neq T_r^u$, then there exists a red edge $wv \in E(T_r^u)$ such that $x'x''$ is reachable from $v$ and not reachable from $w$. Moreover, we can pick red edge $wv \in E(T_r^u)$, such that for all the proper ancestors $v'$ of $v$ in $T_r^u$, $x'x''$ is not reachable from $v'$; and $x'$ is not in the component of $T_r^u - v$ containing $w$.*

**Proof.** First, we shall prove that $x'x''$ is not reachable from $u$. If $j = 0$, then $u = s$. Since $s$ is the common root for all the $k$ edge-disjoint spanning trees $T_1, \ldots, T_k$ in $G^\sigma$; there is no edge (in $G^\sigma$) which is reachable from $s$. If $j \geq 1$, note that if $x'x''$ is reachable from $u$, then by the Growing Algorithm, $x'x''$ is reachable from some vertex in $\vec{H}^u$; but this contradicts with the assumption.

Since $x'x''$ is reachable from $v_0 \in V(T_r^u)$ and not reachable from $u$, there exists some red edge $wv \in E(T_r^u)$ on the path $uT_r^u wvT_r^u v_0$ such that $x'x''$ is reachable from $v$, not reachable from $w$. Suppose that red edge $wv \in E(T_r^u)$ is the first edge on the path $uT_r^u v_0$ with this property; then from any proper ancestor $v'$ of $v$ in $T_r^u$, $x'x''$ is not reachable from $v'$.

To show that $x'$ is not in the component of $T_r^u - v$ containing $w$, it suffices for us considering that $x' = v_0 \in V(T_r^u)$. Note that $v$ is an ancestor of $x'$ in $T_r^u$, therefore $x'$ is not in the component of $T_r^u - v$ containing $w$. $\square$

By Fact 2.7, for the unsaturated directed blue edge $x'x''$ which is reachable from some vertex in $V(T_r^u)$, there is a *special* red edge $wv \in E(T_r^u)$ corresponding to $x'x''$; we say that $x'x''$ *is* 1*-red-reachable from* $wv$ (with respect to $T_r^u$ in $G^\sigma$).

**Lemma 2.8.** *Assume that at time $j$, an unsaturated directed blue edge $x'x''$ is 1-red-reachable from $wv$ (with respect to $T_r^u$ in $G^\sigma$), and $T_r^{x''} \neq T_r^u$. Then $\bigcup_1^k E(T_i) - x'x'' + wv$ has a decomposition $\mathbb{F}' = (T_1', \ldots, T_k', F_{k+1}') \in \mathcal{F}_{k+1}$, satisfying that*

1. *$T_i'$ $(i = 1, \ldots, k)$ are spanning trees of $G$; $\mathbb{F}'$ and its corresponding red-blue-coloring $G^{\sigma'}$ are optimal;*
2. *$T_r^u$ is smaller in $G^{\sigma'}$ than in $G^\sigma$;*
3. *$\sigma' \in \Gamma_j$, i.e., $\vec{H}^u$ has been kept the same in $G^{\sigma'}$ as in $G^\sigma$;*
4. *Moreover, the property that there is no unsaturated directed edge (with respect to $\vec{H}^u$) which is reachable from any vertex $y \in V(\vec{H}^u)$ is kept true in $G^{\sigma'}$.*

**Proof.** Suppose that blue directed path $P := x_0x_1 \ldots x_t$ from $v = x_0$ to $x_t = x''$ (and $x_{t-1} = x'$) witnesses that $x'x''$ is 1-red-reachable from $wv$.

Applying Lemma 2.3 (where $e = x'x''$ and $e' = wv$), denote by $T_1', T_2', \ldots, T_k'$ the trees obtained from $T_1, T_2, \ldots, T_k$, and let $F' = F - wv + x'x''$. Since $T_r^{x'} \neq T_r^{x''}$, $F'$ is a forest. As $x'$, $x''$ are unsaturated in $F$, $\sum_{v \in V(G)} \max\{0, d_{F'}(v) - 1\} \leq \sum_{v \in V(G)} \max\{0, d_F(v) - 1\}$. Hence $\mathbb{F}' = (T_1', T_2', \ldots, T_k', F')$ and its corresponding red-blue-coloring $G^{\sigma'}$ are optimal.

Note that $T_r^u$ in $G^{\sigma'}$ is the component of $T_r^u - v$ containing $w$ in $G^\sigma$. Since $T_r^{x''} \neq T_r^u$, and $x'$ is not in the component of $T_r^u - v$ containing $w$ in $G^\sigma$, we have $T_r^u$ is smaller in $G^{\sigma'}$ than in $G^\sigma$.

Recall we have assume that, for all the vertices $y \in V(\vec{H}^u)$, there is no unsaturated directed edge which is reachable from $y$ with respect to $\vec{H}^u$ in $G^\sigma$; we concluded that for any blue directed path $P_v$ from $v$ to $x'x''$, there are no vertices on $P_v$ belong to $V(\vec{H}^u)$. Therefore, no blue edges in $E(\vec{H}^u)$ have been reoriented. This proves that $\sigma' \in \Gamma_j$, i.e., all the edges in $E(\vec{H}^u)$ have been kept the same in $G^{\sigma'}$ as in $G^\sigma$. And note that the property that there is no unsaturated directed edge which is reachable from any vertex $y \in V(\vec{H}^u)$ with respect to $\vec{H}^u$ is kept true in $G^{\sigma'}$. □

**Lemma 2.9.** *Assume that at time $j$, $T_r^u$ is the red subtree of $G^\sigma$ with root $u$, edge $e = xy$ is on the path $uT_r^uxy$, and there exists $i \in [k]$, such that $x' \in N_i^+(x) \setminus (V(\vec{H}^u) \cup V(T_r^u))$, $y' \in N_i^+(y) \setminus (V(\vec{H}^u) \cup V(T_r^u))$, both $x'$ and $y'$ are unsaturated vertices. Then,*

1. *either edge $xx'$ or $yy'$ can be replaced by the red edge $xy$, resulting in a new decomposition $\mathbb{F}' = (T_1', \ldots, T_k', F_{k+1}') \in \mathcal{F}_{k+1}$, such that $T_i'$ $(i = 1, \ldots, k)$ are spanning trees of $G$, $\mathbb{F}'$ and its corresponding red-blue-coloring $G^{\sigma'}$ are optimal;*
2. *if $yy'$ is replaced by red edge $xy$, then $T_r^u$ is smaller in $G^{\sigma'}$ than in $G^\sigma$; otherwise, $xx'$ is replaced by red edge $xy$, then in $G^{\sigma'}$, $yy'$ is an unsaturated directed blue edge (with respect to $\vec{H}^u$), and is reachable from $x \in V(T_r^u)$.*
3. *$\sigma' \in \Gamma_j$, i.e., $\vec{H}^u$ has been kept the same in $G^{\sigma'}$ as in $G^\sigma$.*

**Proof.** Suppose that $xT_i^\sigma s$ and $yT_i^\sigma s$ firstly meet at vertex $z$, i.e., $(xT_i^\sigma z, zT_i^\sigma y, x)$ is the unique cycle of $T_i + e$; denote this cycle as $C_e$. Then either $e' := (x, x') \in C_e$ or $e' := (y, y') \in C_e$. Let $T_i' := T_i + e - e'$; $T_j' := T_j$ for all $j \neq i$ and $j \in [k]$; then $T_i'$ $(i = 1, \ldots, k)$ are again $k$ edge-disjoint spanning trees of $G$. Let $F_{k+1}' = F_{k+1} + e' - e$; since $e = xy \in E(T_r^u)$, $x', y' \notin V(T_r^u)$, $F_{k+1}'$ is still a forest.

Since both $x'$ and $y'$ are unsaturated vertices of $G^\sigma$, we have $d_{F_{k+1}'}(x') \leq d_{F_{k+1}}(x') + 1 \leq 1$ and $d_{F_{k+1}'}(y') \leq d_{F_{k+1}}(y') + 1 \leq 1$, also $d_{F_{k+1}'}(x) \leq d_{F_{k+1}}(x)$ and $d_{F_{k+1}'}(y) \leq d_{F_{k+1}}(y)$, therefore $\mathbb{F}'$ and its corresponding red-blue-coloring $G^{\sigma'}$ are optimal.

If $yy'$ is replaced by red edge $xy$; in $G^{\sigma'}$, $yy'$ is recolored as red, $xy$ is recolored as blue and oriented from $y$ to $x$. Since $y' \notin V(T_r^u)$, $T_r^u$ is smaller in $G^{\sigma'}$ than in $G^\sigma$. Otherwise, $x' \neq y'$ and $xx'$ is replaced by red edge $xy$; in $G^{\sigma'}$, $xx'$ is recolored as red, $xy$ is recolored as blue and oriented from $x$ to $y$, and $y$ becomes unsaturated; combining $y' \in N_i^+(y) \setminus V(\vec{H}^u)$, both $y$ and $y'$ are unsaturated in $G^{\sigma'}$; therefore, $yy'$ is an unsaturated directed blue edge (with respect to $\vec{H}^u$ in $G^{\sigma'}$), and is reachable from $x \in V(T_r^u)$.

Since $x', y' \notin V(\vec{H}^u)$, $xx', yy' \notin E(\vec{H}^u)$; combining edge $xy \notin E(\vec{H}^u)$, we know that all the edges in $E(\vec{H}^u)$ have been kept the same in $G^{\sigma'}$ as in $G^\sigma$; therefore $\sigma' \in \Gamma_j$. $\square$

In Lemma 2.9, we say that the first $k$ trees $T_i'$ $(i = 1, \ldots, k)$ has been obtained from $T_i$ $(i = 1, \ldots, k)$ by the *replacement* of the edge $e'$ with $e$; before this is done, we say that the edge $e'$ is *replaceable* (by the red edge $e$).

**Lemma 2.10.** *At any time $j$ of the Growing Algorithm, for any optimal orientation $\sigma \in \Gamma_{j+1}$, all the vertices $u \in P$ satisfy Property $P(u)$ with respect to $\vec{H}$ in $G^\sigma$.*

**Proof.** The proof is by induction on time $j$ that vertex $w$ joins $P$ in the Growing Algorithm. Suppose that, at time $j$, just after $w$ joined $P$, optimal orientation set $\Gamma_{j+1}$, vertex set $P$, and subgraph $\vec{H}$, are as defined in the Growing Algorithm.

**Base step ($j = 0$)**: The Growing Algorithm begins with picking $\sigma_0 \in \Gamma_0$, such that $T_r^s$ in $G^{\sigma_0}$ is minimal among all $\sigma \in \Gamma_0$; and $P := \{s\}$.

First we show that, for any optimal orientation $\sigma \in \Gamma_{j+1} = \Gamma_1$, $s$ satisfies (1) of Property $P(s)$. Suppose not, then there exists an unsaturated directed blue edge $x'x''$ in $G^\sigma$, such that $T_r^{x''} \neq T_r^s$, and $x'x''$ is reachable from $v \in V(T_r^s)$. By Fact 2.7 and Lemma 2.8, there exists optimal orientation $\sigma' \in \Gamma_0$, and $T_r^s$ is smaller in $G^{\sigma'}$ than in $G^\sigma$. If the red degree of $s$ is reduced, then this contradicts that $\sigma$ is an optimal orientation of $G$; otherwise, it contradicts that we have picked $\sigma_0 \in \Gamma_0$, such that $T_r^s$ in $G^{\sigma_0}$ is minimal among all optimal orientations $\sigma \in \Gamma_0$.

Next we show that $s$ satisfies (2) of Property $P(s)$. Suppose not, then there exists a red edge $y_1y_2 \in E(T_r^s)$, and $i \in [k]$, such that $y_1' \in N_i^+(y_1) \setminus V(T_r^s)$, $y_2' \in N_i^+(y_2) \setminus V(T_r^s)$, both $y_1'$ and $y_2'$ are unsaturated. Suppose the unique path from $s$ to $y_1y_2$ in $T_r^s$ is $sT_r^s y_1y_2$; note that $s \neq y_1, y_2$. By Lemma 2.9, either edge $y_2y_2'$ or $y_1y_1'$ can be replaced by the red edge $y_1y_2$.

If $y_2y_2'$ is replaceable by red edge $y_1y_2$, then by Lemma 2.9, $T_r^s$ is smaller in $G^{\sigma'}$ than in $G^\sigma$. Since $s \neq y_1$, then $\sigma' \in \Gamma_0$; but this contradicts that we have picked $\sigma := \sigma_0 \in \Gamma_0$ such that $T_r^s$ in $G^{\sigma_0}$ is minimal among all optimal orientations $\sigma \in \Gamma_0$.

If $y_1y_1'$ is replaceable by red edge $y_1y_2$, then by Lemma 2.9, there exists $\sigma' \in \Gamma_0$, in $G^{\sigma'}$, $y_2y_2'$ is an unsaturated directed blue edge, and is reachable from $y_1 \in V(T_r^s)$. Apply Fact 2.7 and Lemma 2.8, suppose the resulting optimal orientation is $\sigma''$. Note that the $T_r^s$ in $G^{\sigma''}$ is contained in the component of $T_r^s - y_1$ containing $s$ in $G^{\sigma'}$; therefore $T_r^s$ is smaller in $G^{\sigma''}$ than in $G^\sigma$; a contradiction.

Note that initially, $P := \{s\}$, $E(\vec{H}) := E(T_r^s) \bigcup_{x \in V(T_r^s)} E_b^+(x)$. Derived from that $s$ satisfies (1) of Property $P(s)$, we have for any vertex $y \in V(\vec{H})$, there is no unsaturated directed blue edge which is reachable from $y$ (in $G^\sigma$).

**Induction hypothesis** ($j \geq 0$): Suppose $u$ joined $P$ at time $j \geq 0$. Suppose $\Gamma_{j+1}$ and $P$ are as defined from the Growing Algorithm, $E(\vec{H}) := E(\vec{H}) \cup E(T_r^u) \bigcup_{x \in V(T_r^u)} E_b^+(x)$, $\vec{H} := G^\sigma[E(\vec{H})]$, all the vertices $x \in P$ satisfy Property $P(x)$ for any optimal orientation $\sigma \in \Gamma_{j+1}$. Moreover, suppose that for the subgraph $\vec{H}$ of $G^\sigma$, for all the vertices $y \in V(\vec{H})$, there is no unsaturated directed edge (with respect to $\vec{H}^u$ in $G^\sigma$) which is reachable from $y$.

**Induction step** ($j \geq 1$): We prove the induction step by contradiction. Suppose $z$ joined $P$ at step $j \geq 1$ of the Growing Algorithm, and $z$ is the first vertex of $P$ in the algorithm that does not satisfy this lemma, and $f(z) = u$. Suppose $P$, $\Gamma_{j+1}$ and $\vec{H}$ are as defined in the algorithm, and there exists some optimal orientation $\sigma \in \Gamma_{j+1}$, such that vertex $z$ in $P$ does not satisfy Property $P(z)$ with respect to $\vec{H}$ in $G^\sigma$. By the definition of Property $P(z)$, we shall consider the time just before $z$ joined $P$.

Informally, except the first vertex $s$, the vertices in $P$ can be regarded as joined to $P$ by the growing process of its father. In the following, according to the different ways that $z$ joined to $P$, the proof of the induction step is divided into two cases.

**Case (i): Suppose $z = w$, $w$ joined $P$ at Line $(i)$ of the Growing Algorithm.**

Since $f(w) = u$, by the algorithm, we have $u \in P$, $w \in N_b^+(v)$, and $v \in V(T_r^u)$. Suppose $u$ joined $P$ at Step $j'$, then $j' < j$.

First we show that $w$ satisfies (1) of Property $P(w)$ for any optimal orientation $\sigma \in \Gamma_{j+1}$. Suppose not, then in $G^\sigma$, there exists unsaturated directed blue edge $x'x''$ with respect to $\vec{H}^w$ (in $G^\sigma$), such that $T_r^{x''} \neq T_r^w$, and $x'x''$ is reachable from some vertex $x \in V(T_r^w)$; $w$ is special in $T_r^w$, since $w \in P$ and we designate $w$ as the root of $T_r^w$.

If $x = w$, since $w \in N_b^+(v)$, $x'x''$ is reachable from $v \in V(T_r^u)$. Note that $u$ joined $P$ at Step $j'$, $j' < j$; $\sigma \in \Gamma_{j+1} \subseteq \Gamma_{j'+1}$. If there exists some vertex in a blue directed path from $v$ to $x'$ belongs to $V(\vec{H}^u)$, then this contradicts the induction hypothesis that, at step $j'$, for all the vertices $y \in V(\vec{H}^u)$, there is no unsaturated directed edge which is reachable from $y$ with respect to $\vec{H}$ (in $G^\sigma$); otherwise, there is a blue directed path $P_v$ from $v$ to $x'$ such that all the vertices in $P_v$ do not belong to $V(\vec{H}^u)$, but this contradicts the induction hypothesis that, at step $j'$, $u$ satisfies (1) of Property $P(u)$ in $G^\sigma$.

Next assume that $x'x''$ is reachable from $x \in V(T_r^w)$, but not from $w$; then it is easy to check that we can apply Fact 2.7 and Lemma 2.8; and we derive that there exists an optimal orientation $\sigma' \in \Gamma_j$ such that $T_r^w$ is smaller in $G^{\sigma'}$ than in $G^\sigma$; but this contradicts that we have picked $\sigma := \sigma_j \in \Gamma_j$, such that $T_r^w$ in $G^{\sigma_j}$ is minimal among all optimal orientations $\sigma \in \Gamma_j$.

Next we show that $w$ satisfy (2) of Property $P(w)$ for any optimal orientation $\sigma \in \Gamma_{j+1}$. Suppose not, then there exists $y_1y_2 \in E(T_r^w)$ in $G^\sigma$, $i \in [k]$, such that $y_1' \in N_i^+(y_1) \setminus (V(\vec{H}^w) \cup V(T_r^w))$, $y_2' \in N_i^+(y_2) \setminus (V(\vec{H}^w) \cup V(T_r^w))$, and both $y_1'$ and $y_2'$ are unsaturated. Suppose the unique path from $w$ to the red edge $y_1y_2$ in $T_r^w$ is $wT_r^wy_1y_2$. Since $w$ joined $P$ at this step, we know that at the time just before $w$ joined $P$, $V(T_r^w) \cap P = \emptyset$. By Lemma 2.9, either edge $y_2y_2'$ or $y_1y_1'$ can be replaced by red edge $y_1y_2$.

If $y_2y_2'$ is replaceable by red edge $y_1y_2$, then by Lemma 2.9, then there exists $\sigma' \in \Gamma_j$, such that $T_r^w$ is smaller in $G^{\sigma'}$ than in $G^\sigma$; this contradicts that we have picked $\sigma := \sigma_j \in \Gamma_j$, such that $T_r^w$ in $G^{\sigma_j}$ is minimal among all optimal orientations $\sigma \in \Gamma_j$.

If $y_1y_1'$ is replaceable by red edge $y_1y_2$, then by Lemma 2.9, there exists $\sigma' \in \Gamma_j$, in $G^{\sigma'}$, $y_2y_2'$ is an unsaturated directed blue edge (with respect to $\vec{H}^w$), and is reachable from $y_1 \in V(T_r^w)$. By the induction hypothesis, for any vertex $y \in V(\vec{H}^w)$, there is no unsaturated directed edge which is reachable from $y$ with respect to $\vec{H}$; we concluded that $y_2y_2'$ is not reachable from $w$. Apply Fact 2.7 and Lemma 2.8, suppose the resulting optimal orientation is $\sigma''$. Note that the new $T_r^w$ in $G^{\sigma''}$ is contained in the component of $T_r^w - y_1$ containing $w$ in $G^{\sigma'}$; therefore $T_r^w$ is smaller in $G^{\sigma''}$ than in $G^\sigma$; since $\sigma'' \in \Gamma_j$, this is a contradiction.

**Case (ii): Suppose $z = q$, $q$ joined $P$ at Line $(ii)$ of the Growing Algorithm.**

The proof for this case is similar to **Case (i)**, the details are skipped here. □

**Proof of Theorem 1.6.** The Growing Algorithm returns a set of optimal orientations $\Gamma^* \neq \emptyset$ of $G$, a vertex set $P^*$, and a subgraph $\vec{H}^* := G^{\sigma^*}[E(\vec{H}^*)]$ of $G^{\sigma^*}$. We shall show the subgraph $\vec{H}^*$ of $G^{\sigma^*}$ returned does not satisfy the condition of Theorem 1.6, hence the counterexample $G$ of Theorem 1.6 does not exist, and the theorem holds.

Now for all the running time of the Growing Algorithm, fix the referring optimal orientation $\sigma$ being an optimal orientation $\sigma^* \in \Gamma^*$. By Lemma 2.10, at any time of the algorithm, all the vertices $u \in P$ satisfy property $P(u)$ with respect to $\vec{H}$ in $G^{\sigma^*}$.

**First we count the number of blue edges in $\vec{H}^*$.**

Recall that initially, $E(\vec{H}) := E(T_r^s) \bigcup_{x \in V(T_r^s)} E_b^+(x)$ and $d_r(s) > 1$, $\vec{H} := G^{\sigma^*}[E(\vec{H})]$. Whenever a new vertex $p$ joined $P$ in the algorithm, $\vec{H}$ is updated as following: $E(\vec{H}) := E(\vec{H}) \cup E(T_r^p) \bigcup_{x \in V(T_r^p)} E_b^+(x)$, $\vec{H} := G^{\sigma^*}[E(\vec{H})]$. Therefore, for any vertex $x \in V(\vec{H})$, if $x \neq s$ and $x \in V(T_r^p)$ (where $p \in P$), the blue edges $E_b^+(x)$ are included in $E(\vec{H})$.

If $x \in V(\vec{H})$, $x \neq s$, and $x \notin V(T_r^p)$ (where $p \in P$), then $x \in N_b^+(v)$ for some $v \in V(T_r^q)$ where $q \in P$. By the algorithm, either $x \in V(T_r^p)$ for some $p \in P$ (this case has been considered already), or $x$ is an unsaturated vertex (*we only need to consider this*

*case*). By Lemma 2.10, all the vertex $p \in P$ satisfy Property $P(p)$; by $(1)'$ of Property $P(p)$ and the Growing Algorithm, all the blue edges $E_b^+(x)$ are included in $E(\vec{H})$.

Since $\sigma^*$ is an optimal orientation, $|E_b^+(x)| = k$ for all $x \neq s$. Therefore $E(\vec{H}^*)$ contains $k(|V(\vec{H}^*)| - 1)$ blue edges.

**Next we count the number of red edges in $\vec{H}^*$.**

Define $\overline{P} := \bigcup_{u \in P^*} V(T_r^u)$; here, $P^*$ is the final $P$ returned by the Growing Algorithm. Next, by induction on $|P|$ (in the algorithm), we show that $V(\vec{H}^*)$ can be covered by $\bigcup_{u \in P^*} P'_u$, where $P'_u = V(T_r^u) \cup (\bigcup_{v \in V(T_r^u)} N_b^+(v) \setminus \{V(\vec{H}^u) \cup \overline{P}\})$. Note that $\bigcup_{u \in P^*} P'_u = \overline{P} \bigcup_{u \in P^*} P'_u$.

Base step, $|P| = 1$, then $P = \{s\}$, and $E(\vec{H}) := E(T_r^s) \bigcup_{x \in V(T_r^s)} E_b^+(x)$, $V(\vec{H}^s) = \emptyset$. Then by the Growing Algorithm and $V(\vec{H}^s) = \emptyset$,

$$\begin{aligned} V(\vec{H}) &:= V(T_r^s) \bigcup_{x \in V(T_r^s)} N_b^+(x) \subseteq \overline{P} \cup (\bigcup_{x \in V(T_r^s)} N_b^+(x) \setminus \overline{P}) \\ &= \overline{P} \cup (\bigcup_{x \in V(T_r^s)} N_b^+(x) \setminus \{V(\vec{H}^s) \cup \overline{P}\}) \subseteq \overline{P} \bigcup_{u \in P} P'_u \subseteq \bigcup_{u \in P^*} P'_u. \end{aligned}$$

Induction step, whenever a new vertex $q$ joined $P$ in the Growing Algorithm, $E(\vec{H})$ is updated by $E'(\vec{H}) := E(\vec{H}) \cup E(T_r^q) \bigcup_{x \in V(T_r^q)} E_b^+(x)$. Note here, by the Growing Algorithm, for the newly added vertex $q$, we have $V(T_r^q) \cap P = \emptyset$, therefore $E(T_r^q) \cap (\bigcup_{u \in P} E(T_r^u)) = \emptyset$, where $P$ and $\vec{H}$ are as defined by the induction hypothesis; this fact is used later in the counting of the number of red edges in $\vec{H}$. By the induction hypothesis, $V(\vec{H}) \subseteq \bigcup_{u \in P^*} P'_u$.

For the newly added vertex $q$, $P'_q = V(T_r^q) \cup (\bigcup_{v \in V(T_r^q)} N_b^+(v) \setminus \{V(\vec{H}^q) \cup \overline{P}\})$, then

$$V'(\vec{H}) := V(\vec{H}) \cup V(T_r^q) \bigcup_{x \in V(T_r^q)} N_b^+(x) \subseteq (\bigcup_{u \in P^*} P'_u) \cup (\overline{P} \cup P'_q) = \overline{P} \bigcup_{u \in P^*} P'_u = \bigcup_{u \in P^*} P'_u.$$

This finishes the induction step. We concluded that the family of sets $\mathcal{P} := \cup\{P'_u : u \in P^*\} = P'_s \cup \{P'_u : u \in P^* \setminus \{s\}\}$ covers $V(\vec{H}^*)$.

**Claim 2.11.** *For each vertex $u \in P^*$, $\frac{|E(T_r^u)|}{|P'_u|} \geq \frac{1}{k+2}$. Moreover, if $u = s$, $\frac{|E(T_r^s)|}{|P'_s|} > \frac{1}{k+2}$.*

**Proof.** For vertex $u \in P^*$, all the vertices $x \in V(T_r^u)$ are saturated (having $d_r(x) \geq 1$), and $|V(T_r^u)| \geq 2$. Suppose $wv \in E(T_r^u)$, by (2) of Property $P(u)$ and the Pigeonhole Principle, $|\bigcup_{x \in \{w,v\}} \{N_b^+(x) \setminus (V(\vec{H}^u) \cup \overline{P})\}| \leq k$; for all the vertices $x \in V(T_r^u) \setminus \{w, v\}$, due to $d_b^+(x) = k$, $|\{N_b^+(x) \setminus (V(\vec{H}^u) \cup \overline{P})\}| \leq k$; therefore, $|\bigcup_{x \in V(T_r^u)} \{N_b^+(x) \setminus (V(\vec{H}^u) \cup \overline{P})\}| \leq (|V(T_r^u)| - 1)k$. Since $|E(T_r^u)| = |V(T_r^u)| - 1$, we have

$$\frac{|E(T_r^u)|}{|P'_u|} = \frac{|E(T_r^u)|}{|V(T_r^u) \bigcup_{x \in V(T_r^u)} \{N_b^+(x) \setminus (V(\vec{H}^u) \cup \overline{P})|}$$
$$\geq \frac{|V(T_r^u)| - 1}{|V(T_r^u)| + (|V(T_r^u)| - 1)k} \geq \frac{1}{k+2}.$$

In particular, if $u = s$, since $d_r(s) > 1$, $|V(T_r^s)| > 2$, we have $\frac{|E(T_r^s)|}{|P'_s|} > \frac{1}{k+2}$. $\square$

By Claim 2.11, for $u \in P^*$, $P'_u$ contains at least $\frac{1}{k+2}|P'_u|$ red edges, $P'_s$ contains more than $\frac{1}{k+2}|P'_s|$ red edges, combining $\mathcal{P} = P'_s \cup \{P'_u : u \in P^* \setminus \{s\}\}$ covers $V(\vec{H}^*)$, thus the number of red edges in $\vec{H}^*$ is larger than $\frac{1}{k+2}|V(\vec{H}^*)|$, plus the number of blue edges in $\vec{H}^*$, we have $e(\vec{H}^*) > k(v(\vec{H}^*) - 1) + \frac{1}{k+2}v(\vec{H}^*)$. On the other hand, by the assumption of Theorem 1.6, $e(\vec{H}^*) \leq k(v(\vec{H}^*) - 1) + \frac{1}{k+2}v(\vec{H}^*)$, a contradiction. This finishes the proof of Theorem 1.6. $\square$

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